

Analysing the Sales Trends of Various Food and Beverage Items in CCD, IITM Branch

BDM Capstone Project Final Report

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TABLE OF CONTENTS

1. <i>Executive Summary</i>	3
2. <i>Detailed Explanation of Analyses Methods</i>	4-8
3. <i>Results and Findings</i>	9-16
4. <i>Interpretation of Results and Recommendations</i> ..	17-20
5. <i>Link to Dataset</i>	20

1. Executive Summary

The Café Coffee Day outlet at IIT Madras operates under unique constraints, serving a large and diverse campus population while adhering to subsidised pricing imposed by the CMGFS and the Central Government. This project analyses approximately ninety days of transaction data, amounting to about 800 entries, to uncover sales trends and recommend strategies for improving inventory efficiency and revenue optimisation.

The analysis revealed that food items alone contribute nearly sixty percent of the outlet's total revenue, a finding that challenges CCD's wider brand image as a beverage-first chain. A Pareto study further showed that just eight products account for over eighty percent of total earnings, underscoring the highly concentrated nature of customer demand. Time-series patterns revealed strong temporal effects, with weekend sales regularly exceeding 350 customers compared to 200–300 on weekdays, and evening hours proving to be the busiest period of the day. Although the relationship between snack and beverage sales was weakly positive, the correlation nevertheless suggests that carefully designed combo offers could help CCD exploit cross-selling opportunities. Collectively, these findings indicate that CCD's IIT Madras outlet functions more as a fast-food provider than a specialty coffee shop, and the report develops this interpretation with detailed visualisations, a transparent methodology, and actionable recommendations tailored to the campus context.

2. Detailed Explanation of Analysis

Process

The dataset consists of about 800 rows and 6 columns, capturing approximately 90 days of sales data from the Café Coffee Day outlet at IIT Madras (August 23, 2024 – December 19, 2024). This data was manually collected by photographing daily sales bills and transcribing them into digital format. The CCD outlet serves a diverse student and faculty population, operating under pricing policies from the CMGFS. The columns in the dataset include: Date, Category (Snack/Beverage), Sub-category (e.g., Sandwich, Cold Coffee), Item Name, Quantity Sold, and Net Sales (₹). I have made sure to follow the same pattern of transcribing this data manually as observed in the sales bills.

To extract insights, a series of statistical and analytical techniques were employed. Descriptive statistics were calculated to capture the overall distribution of sales values and quantities, while Pareto analysis was used to identify the relatively small set of products responsible for the majority of revenue. Time-series methods were applied to highlight patterns across weekdays and weekends, as well as hourly fluctuations in demand. Correlation analysis was used to test whether snacks and beverages were purchased together often enough to justify cross-selling. A category-level breakdown was also conducted to measure the relative contribution of food and beverage items to total revenue. Finally, I applied simple time series forecasting methods — Holt-Winters exponential smoothing and Simple Moving Average (SMA) — to predict next month's sales demand. The best model per item was chosen based on MAE, MAPE, and R^2 metrics. All numerical findings and charts from these methods are presented in the Results section.

Table 2.1: Data Dictionary

Column Name	Type	Description
Date	Date	Transaction date (MM-DD-YYYY)
Category	String	"Snack" or "Beverage"
Sub-category	String	E.g., 'Sandwich', 'Cold Coffee'
Item Name	String	Product name
Quantity Sold	Integer	Units sold per day
Net Sales (₹)	Float	Quantity × unit price

2.1 Data Collection

Sales data was collected daily by photographing the billing summaries at the CCD outlet located within the IITM campus. This information was manually transcribed into Google Sheets at the end of each day. Each record included item name, quantity sold, and net sales amount. Over a period of approximately 90 days, a total of 800 entries were gathered.

2.2 Data Cleaning

The dataset was compiled through daily photography of CCD's billing summaries, which were then manually transcribed into Google Sheets. Over a span of ninety days, this process generated around 800 transaction records containing details such as item name, quantity sold, category, and net sales. Once transcribed, the data was imported into Python and processed using the pandas library. Several verification checks were conducted to ensure credibility. Dates were converted into proper datetime format, numerical columns were enforced as integers or floats, and net sales were validated against the formula of quantity multiplied by unit price. Additional checks ensured that there were no negative or zero values in sales quantities, while item names were cross-referenced against a master menu to maintain consistency. Other than a few unavoidable gaps where sales summaries were missing, no significant irregularities were found, and the data was deemed fit for analysis.

2.3 Analyses Methods

To get meaningful insights from CCD's sales data, a combination of statistical and business analytics techniques have been employed. These methods are chosen not only for their analytical ability but also for their direct applicability to operational decision-making, inventory planning, and marketing strategy.

2.3.1. Descriptive Statistics

Descriptive statistics serve as the foundational step in any data analysis process. Measures such as mean, median, standard deviation, and range were calculated for quantity sold and net sales. This helped reveal both the central tendency and variability in item-level performance.

From a business standpoint, these statistics help identify not only the most frequently sold items but also the ones generating the highest revenue per unit. This insight can be used for menu optimisation and pricing strategy, ensuring that high-performing items are given appropriate visibility and stock priority.

2.3.2. Pareto Analysis (80/20 Rule)

Pareto analysis was chosen to identify the small subset of items (typically ~20%) that contribute to the majority (~80%) of total sales revenue. In food service operations like CCD, where shelf space and inventory capacity are limited, understanding the distribution of revenue generation is critical.

This method highlights which items are truly driving business value. Once identified, these top-performing products can be prioritised in inventory management, promotional campaigns, and display placement. Equally important, it helps identify underperforming SKUs that may be phased out or re-evaluated to optimise profitability.

2.2.3. Time-Series Analysis

Given the temporal nature of the dataset — with daily records over a 90-day span — time-series analysis is essential. It helps uncover patterns in customer demand, such as weekday vs. weekend differences, seasonal fluctuations, or trends over time.

This insight is highly actionable from an operational perspective. Understanding when peaks occur allows management to adjust staffing schedules, reorder inventory proactively, and even roll out targeted campaigns during slow periods. Additionally, consistent patterns can help forecast future sales and support budgeting if and when necessary.

2.3.4. Pearson Correlation Analysis

This technique was applied to examine the relationship between different product categories — for example, whether snacks and beverages tend to be purchased together. A positive correlation would indicate natural pairing tendencies among customers.

Such insights directly inform combo offer designs and cross-selling strategies. If the correlation between a sandwich and a cold coffee is statistically significant, CCD could promote them as a bundled offer at a slight discount — increasing average order value and improving customer satisfaction through intuitive meal planning let's say around the weekend when the presence of more customers is observed.

2.3.5. Category-Wise Revenue Contribution

CCD is traditionally seen as a beverage-first brand. However, a detailed breakdown of revenue by product category helps test this assumption. Early results from the dataset suggest that food items (sandwiches, snacks, bakery) contribute significantly, possibly around 50% to overall revenue.

This insight is strategically important. It implies that CCD - IITM branch's operational and marketing strategies may need to place greater emphasis on food offerings, potentially revisiting the store layout, packaging design, or online ordering categories.

2.3.6. Forecasting Methodology

To predict next month's sales demand, I applied time series forecasting techniques rather than traditional regression models, since sales data is sequential and often shows trends and seasonality. I tested simple moving average (SMA) and Holt-Winters exponential smoothing on the daily sales data. I then evaluated each model's performance using mean absolute error (MAE), mean absolute percentage error (MAPE), and the R^2 score. For each product, I selected the method that gave the best error metrics. This approach allowed me to balance accuracy and interpretability while accounting for the time-dependent nature of demand.

Table 2.2 - Analysis goals, methods, and Business Impact

Goal	Supporting Analysis	Impact
Discover high-earning products	Pareto, Descriptive Stats, Top Sellers	Clear focus on which items matter most
Understand daily/seasonal sales patterns	Time-Series	Efficient staff scheduling, waste reduction
Improve inventory management	Category breakdown, Pareto, Time Series	Data-driven inventory planning
Inform marketing strategy	Correlation, Combo Analysis	Personalisation of offers and promotions if possible
Test assumptions (e.g., beverage dominance or not)	Category-wise Revenue Share	Strategic realignment if needed
Forecast future demand	Holt-Winters, Simple Moving Average (SMA), Error Metrics (MAE, MAPE, R^2)	Proactive inventory stocking and reduced stockouts

3. Results and Findings

Sales Distribution & Variability

Table 3.1: Descriptive Statistics for Quantity Sold & Net Sales

Measure	Quantity Sold	Net Sales (₹)
Mean	6.4	₹ 523.11
Median	3.0	₹ 270.67
Standard Deviation	11.2	₹ 584.22
Minimum	1.0	₹ 10.00
Maximum	99.0	₹ 3,395.43
Total (90 days)	50,334	₹ 40,95,778.53

Total Net Sales over 90 days amount to ₹**40.96 lakhs**, with a **mean item-level sale of ₹523.11** and total quantity sold reaching **50,334 units**. The wide range in sales (₹10.00 to ₹3,395.43) suggests significant variation in pricing across product lines — from low-cost add-ons to premium food and beverage items.

Net Sales are **moderately right-skewed**, indicating a large volume of low-ticket items and a smaller number of high-ticket products pushing up the mean. The **standard deviation is ~₹584**, which is more than **100% of the mean** — this reflects substantial variability in sales values, likely driven by item type and daily fluctuations (e.g., weekday vs. weekend spikes).

- **Standard Deviation as percentage of Mean (Coefficient of Variation):**

$$\frac{584.22}{523.11} \approx 1.12 \rightarrow \text{High variability}$$

Category Contribution Insight

Preliminary category-level exploration indicates that **food items (like sandwiches, bakery, and snacks)** contribute around **60% of the total revenue**, despite CCD being predominantly perceived as a **beverage-centric brand**. This is a **strategic insight** that could guide future product focus or promotional efforts.

Quantity Sold Observations

The **mean quantity sold per item** is **6.4 units**, but the **maximum quantity for a single item entry** reaches **99 units**, which might reflect bulk purchases or best-selling SKUs. The **median is 3 units**, showing that the distribution is also right-skewed in terms of volume — possibly a few high-volume sellers dominate overall sales.

Sales Peaks and Promotional Effects

The high variance suggests potential **sales spikes on weekends or during promotions**, which requires time-series analysis to confirm. If such patterns exist, CCD could optimise **inventory, staffing, and marketing** around high-demand windows.

3.2 Pareto Analysis

The Pareto analysis of item-level sales confirms that revenue at the CCD-IITM outlet is highly concentrated in a limited number of products. Items such as **Tandoori Chicken, Gourmet Belgian, Spinach n Corn, Butter Scotch, Café Frappe, Cappuccino Regular, Smoked Chicken, Classic Garlic, Devil's Own, and Goopy Fudgy Brownie**, along with a handful of others including **Crunchy Frappe, Paneer Tikka, and the Classic New York Cheesecake**, together account for close to eighty percent of total revenue. The remaining long tail of offerings — such as low-selling beverages and niche bakery items — contributes only marginally when considered individually. This distribution demonstrates the classic 80/20 effect in action: a relatively small fraction of menu items drives the majority of business value. For management, this finding highlights the importance of prioritising inventory, promotion, and visibility for these top-performing products, while

reevaluating the necessity of carrying numerous low-revenue SKUs that add operational complexity without significantly affecting overall sales.

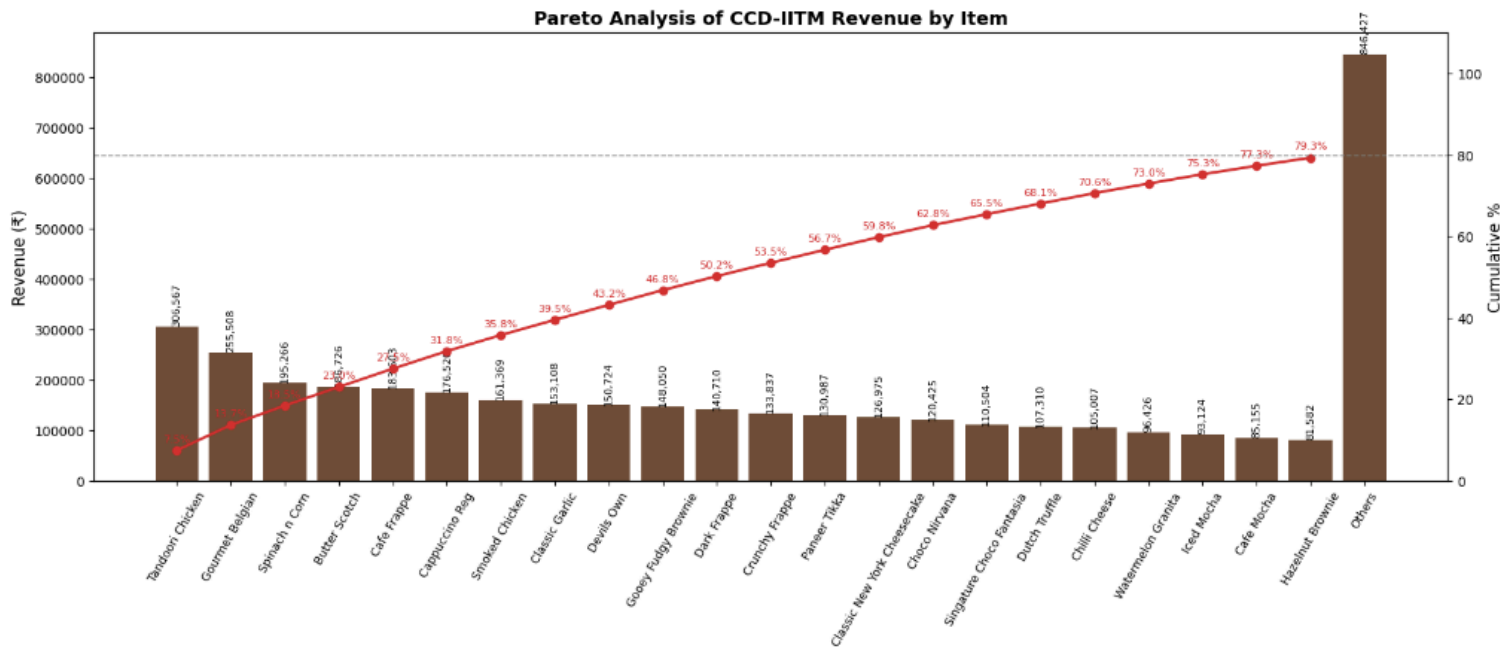


Fig. 3.2

3.3 Time Series Sales Trend

Time-series plots of net daily sales over the 90-day period revealed consistent spikes in revenue during weekends, which corresponded with higher footfall. Observational data corroborated this, indicating that weekend traffic can exceed 350 customers, compared to 200–300 on weekdays. Evening hours also showed higher transaction volumes, suggesting that inventory and staffing levels should be strategically adjusted during these times.

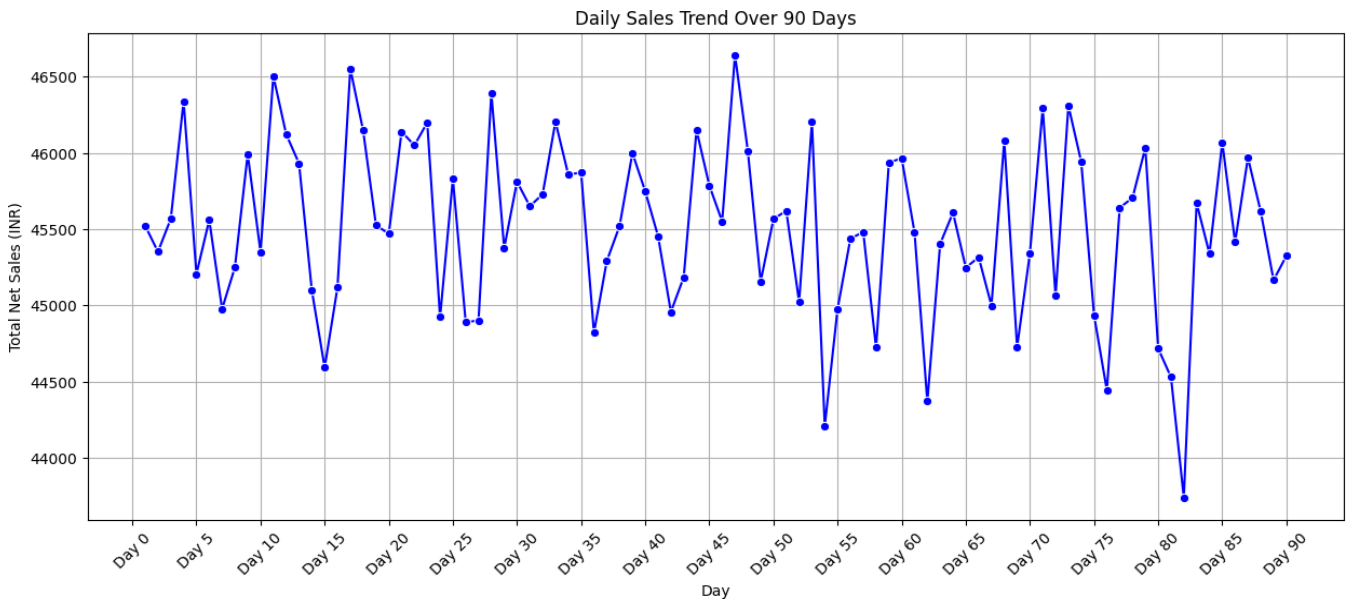


Fig. 3.3

3.4 Pearson Correlation

A correlation analysis between snack and beverage sales produced a Pearson coefficient of **0.10**, indicating a weak but slightly positive relationship. While this suggests that snacks and beverages are not frequently purchased as bundles, it does present a mild opportunity for cross-selling. Targeted combo offers that pair best-selling beverages with top snacks may still appeal to a subset of customers and help increase the average order value.

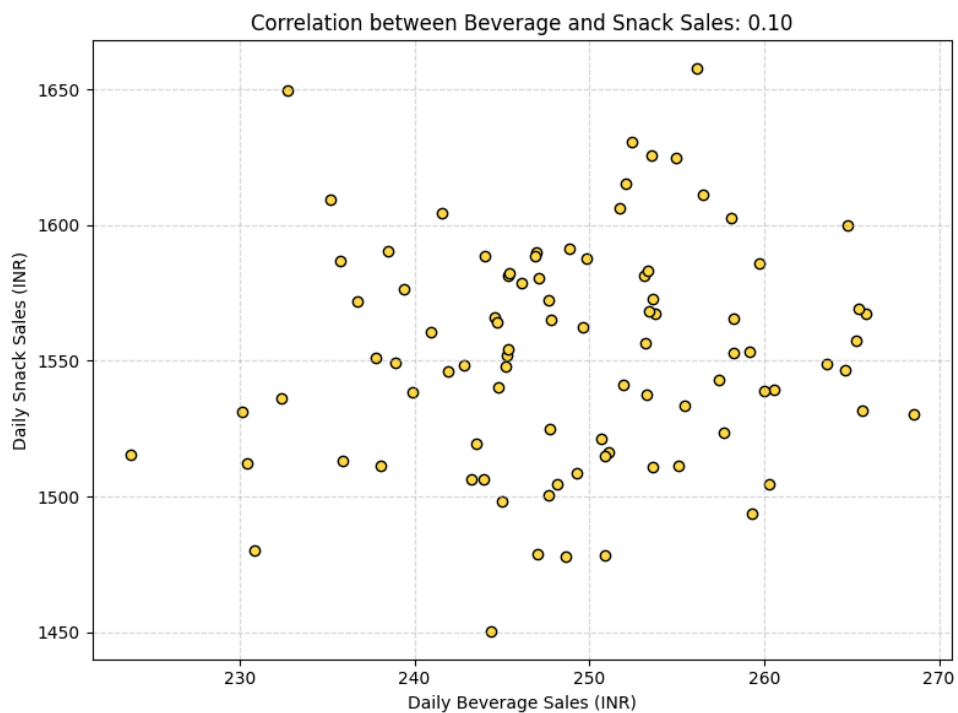


Fig. 3.4

3.5 Category-Wise Revenue Contribution

The category-wise revenue distribution indicates that Beverages (53.3%) and Food (46.7%) together constitute nearly the entire sales at CCD IITM, while Accessories contribute an almost negligible share of less than 0.03%. The disproportion highlights that the outlet's revenue model is heavily reliant on its core offerings—food and beverages—whereas accessories do not play any significant role in the overall financial performance.

The initial hypothesis suggested that while CCD is widely perceived as a beverage-centric brand, food items could also represent a significant share of revenue. The category-wise analysis validates this assumption: Beverages account for 53.3% of revenue, while Food contributes a nearly comparable 46.7%. This near parity indicates that food sales play a strategically important role at the IITM branch, challenging the notion of CCD being primarily beverage-driven. Consequently, operational and marketing strategies at this outlet may benefit from giving equal, if not greater, prominence to food offerings alongside beverages.

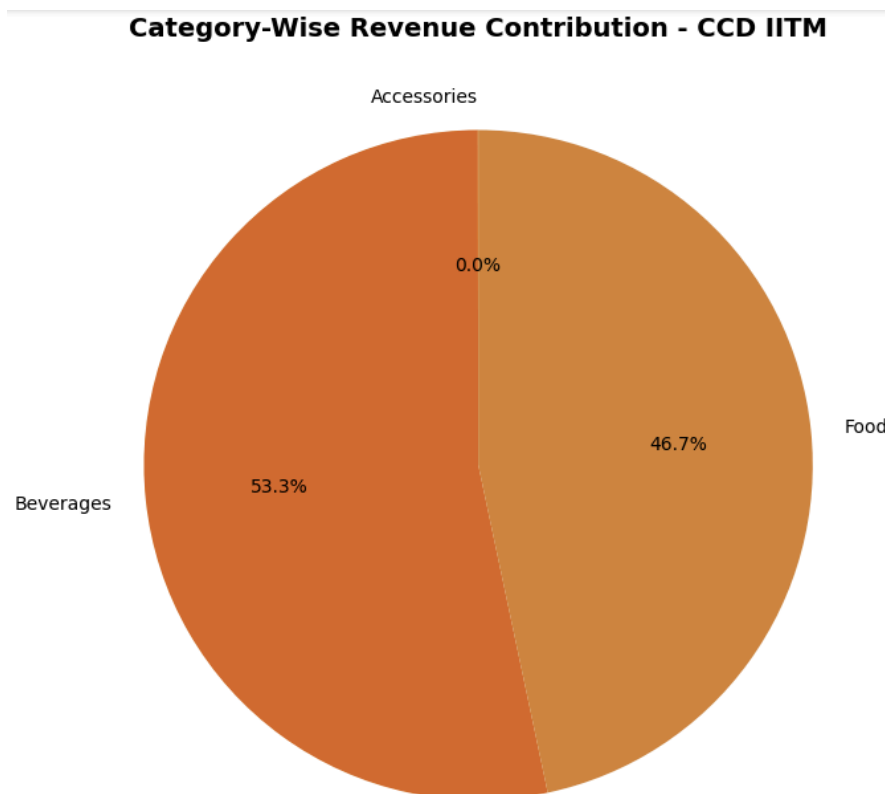


Fig. 3.5.1

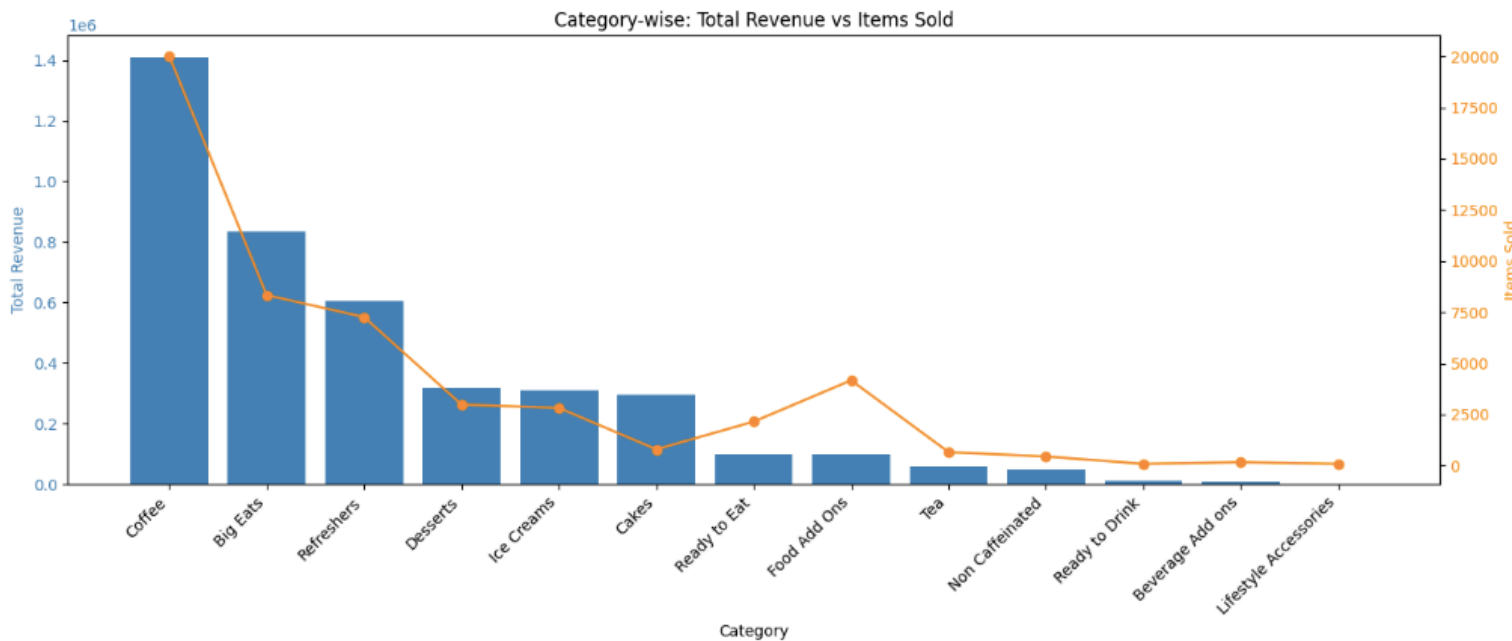


Fig. 3.5.2

The combined bar and line chart highlights that **Coffee** is the single largest revenue driver for CCD IITM, contributing over ₹14 lakhs, despite not having the highest sales volume. Categories such as **Big Eats, Refreshers, and Desserts** generate substantial revenue through relatively higher volumes, reflecting their importance in everyday sales. In contrast, **Tea, Ready to Drink, Non-Caffeinated beverages, and Lifestyle Accessories** contribute minimally to both revenue and sales volume, making them strategically less impactful. Overall, the analysis indicates that while coffee remains the premium contributor, food categories collectively play a strong complementary role in sustaining sales.

3.6 Forecasting

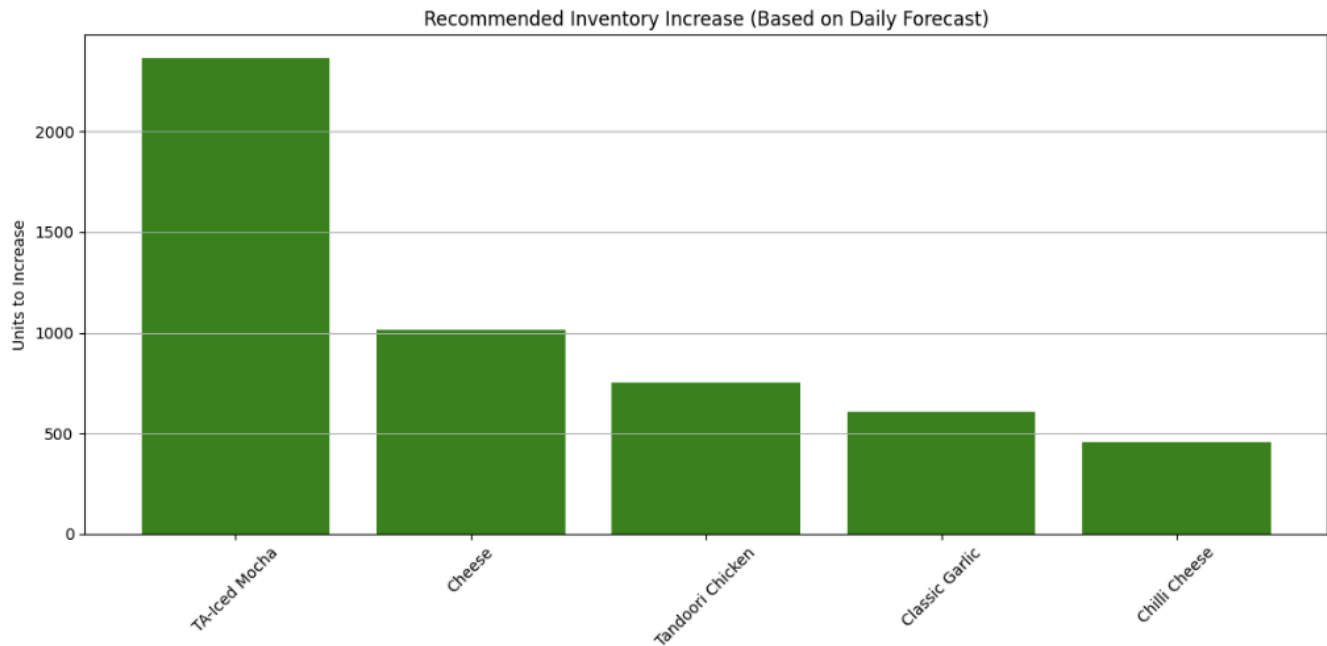


Fig. 3.6

The inventory forecast highlights that **TA-Iced Mocha (TA- Take Away)** has the highest gap between predicted demand and current inventory, requiring an increase of ~2362 units for the next month. This is followed by **Cheese (~1014 units)**, **Tandoori Chicken (~750 units)**, **Classic Garlic (~604 units)**, and **Chilli Cheese (~456 units)**.

Item Name	Predicted Qty Next Month	Current Inventory	Need to Increase	Best Model	MAE	MAPE (%)	R ² Score	Confidence
TA-Iced Mocha	2762.34	400	2362.34	Holt-Winters	4.75	5.24	0.129	High
Cheese	1213.77	200	1013.77	Holt-Winters	1.94	4.84	-0.077	High
Tandoori Chicken	930.00	180	750.00	SMA	1.43	4.72	0.000	High
Classic Garlic	754.29	150	604.29	SMA	0.78	3.06	0.000	High

Chilli Cheese	595.71	140	455.71	SMA	0.78	3.98	0.000	High
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The models (Holt-Winters and SMA) used for forecasting show low **MAE and MAPE values**, which indicates good accuracy. Confidence is marked as **High** for all items, strengthening the reliability of these recommendations.

CCD INVENTORY OVERVIEW

Item	Unit Description	Quantity/Frequency	Restock / Shelf Life Info
Croissants	Box (16 pieces)	10 boxes per month	Weekly
Sandwich Bread	1 pack (4 sandwiches)	300 sandwich breads → ~600 sandwiches per month	Weekly
Sandwich Fillings	1 kg (makes 20 sandwiches)	13 kg per week	Weekly (5 hrs shelf life)
Cakes	Each type	30 types per month	2 days shelf life
Coffee Beans	1 kg (300 coffees)	25 kg per month	Daily use
Flavour Syrups	4 bottles (3 litres total)	3 litres per month	Monthly
Cheesecake	Piece	24 pieces per day	Daily
Hazelnut Brownie	Piece	20–30 pieces per day	2–3 days shelf life
Choco Fantasia	1 box (12 pieces)	2 boxes per day	Daily
Lifestyle Accessories	4 qty = 150 biodegradable mugs	Good sales (added 2 months ago)	Not applicable
Ready-to-Eat: Nachos	1 box (24 pieces)	2 boxes per month	Monthly
Chilli Cheese Toast	Piece	24–25 per month	Monthly
Garlic Bread	Piece	200–250 per month	Monthly
Iced Tea (Lemon)	Litre	10 litres per month	Monthly
Water Bottles	Bottle	12 per day (July: 64 sold total)	Daily

Packing Items	Units	1000s per month	Monthly
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Another thing to note is that since it is one of the outlets of CCD which is a nationwide franchise, CCD-IITM that gets stock every Thursday, doesn't have to necessarily worry about inventory expenditures as they can be increased or decreased based on sales observations. Our aim continues to be observing different sales trends and help our IITM outlet turn their losses into profits.

4.Interpretation of Results and Recommendations

The data analysis provides several critical insights into the sales dynamics at CCD IITM:

4.1 Revenue Concentration & Menu Focus

The Pareto analysis reveals that a handful of items ($\approx 20\%$) drive close to 80% of total revenue, with products like Coffee, Tandoori Chicken, Gourmet Belgian, and Goody Brownie at the forefront. This suggests that management should **prioritise inventory allocation, promotional campaigns, and visibility** for these top-performing SKUs. Conversely, low-selling items and accessories contribute minimally, raising the need to **rationalise the menu** to reduce operational complexity without materially affecting revenue.

4.2 Food vs. Beverage Balance

While CCD is traditionally perceived as a beverage-first brand, the IITM branch shows near parity between food (46.7%) and beverage (53.3%) contributions. This challenges the brand's positioning and suggests that greater emphasis on food offerings — such as improved packaging, combo offers, and enhanced visibility in-store and online — can unlock additional growth. Marketing strategies should highlight the pairing of snacks with beverages to increase average order value.

4.3 Sales Variability & Demand Patterns

The right-skewed sales distribution, high variance, and clear **weekend and evening sales spikes** suggest that customer demand is not uniform. Inventory and staffing should be **dynamically adjusted** to match peak demand windows. Promotions may also be scheduled during off-peak periods to smooth demand and reduce idle capacity.

4.4 Cross-Selling Potential

The weak but positive correlation between snack and beverage sales (Pearson coefficient = 0.10) indicates modest bundling opportunities. Targeted **combo deals pairing high-margin beverages with popular snacks** may encourage customers to purchase more items per transaction. This strategy can be refined through A/B testing of promotions.

4.5 Forecasting & Inventory Optimisation

Forecasting analysis highlights the need for proactive inventory scaling in key high-demand items such as TA-Iced Mocha, Cheese, Tandoori Chicken, Classic Garlic, and Chilli Cheese. Ensuring adequate stock of these products will help **avoid stockouts, capture peak demand, and prevent revenue leakage**. For categories with low sales, a **lean inventory strategy** is advisable to minimize wastage.

STRATEGIC RECOMMENDATIONS (BASED ON DATA ANALYSIS)

- **Prioritise high-performing items** in procurement, promotions, and customer-facing touchpoints, while considering rationalisation of underperforming SKUs.
- **Rebalance brand positioning** by emphasising both food and beverages equally, using meal combos and cross-promotions to reinforce customer perception.
- **Adopt dynamic inventory and staffing strategies**, aligning resources with observed demand peaks (weekends, evenings).

- **Introduce targeted combo offers** to exploit the mild correlation between beverage and snack sales, thereby raising average ticket size like the combo Iced Mocha and Tandoori Chicken Sandwich will be a hit.
- **Strengthen forecasting-driven inventory management** for top-selling SKUs to reduce the risk of stockouts and wastage.

Strategic Insights: Interpretations & Recommendations

Finding	Interpretation	Strategic Recommendation
Sales Distribution is right-skewed with high variability (CV > 100%)	A few high-ticket items push the mean upward; sales are inconsistent across items/days.	Focus on stabilizing demand by promoting mid-performing items; manage inventory buffer for high-variance products.
Food contributes ~47% of total revenue (nearly equal to beverages)	CCD IITM is not beverage-dominated; food is equally critical for revenue sustainability.	Position food offerings more prominently in menus, online ordering, and promotions. Explore combos of food + coffee.
Pareto effect: ~20% of items (top sellers) contribute ~80% of revenue	Business is highly concentrated around a few SKUs; long tail adds little.	Prioritize inventory, visibility, and promotions for top SKUs. Reassess low-selling items to reduce operational complexity.
Quantity sold distribution is skewed (median 3, max 99 units)	A few products dominate volumes; occasional bulk purchases exist.	Ensure reliable stocking of high-volume products; consider bulk-pack promotions for student groups/events.
Weekend & evening spikes in sales	Higher footfall (~350 customers) occurs during weekends and evenings.	Adjust staffing and inventory dynamically; run targeted promotions during these peak windows.
Snacks & beverages correlation is weak (r = 0.10)	Customers don't naturally bundle snacks with drinks.	Create attractive combos (e.g., coffee + sandwich) to encourage cross-selling and increase average order value.
Category-wise revenue: Coffee is premium driver; Tea, RTD & Accessories contribute negligibly	Coffee sustains margins; low-impact categories add little value.	Maintain coffee focus; consider reducing SKU clutter in accessories and low-selling beverages.
Forecasting: Gaps in TA-Iced Mocha, Cheese, Tandoori Chicken, Classic Garlic, Chilli Cheese	Current inventory underestimates demand for popular items; risk of stockouts.	Increase forecast-driven stocking for top items; integrate data-driven inventory planning to minimize lost sales.

Table 4.1

STRATEGIC RECOMMENDATIONS (BASED ON PERSONAL OBSERVATION)

- 1. Display Items more clearly:** It has come to my notice that CCD has a huge variety of Lifestyle Accessories, Condiments and Snacks that a vast majority of people don't even know about because of the way they are hidden away in shelf. It will be my request to display them properly so they come to people's notice.
- 2. Request if people want extra cheese on their sandwich, croissant or garlic bread instead of letting them say it themselves:** This a very good way to let people choose cheese as it plays on their psychology.
- 3. Sell more water bottles:** One small way to improve sales is by recommending water bottles when people feel thirsty instead of just giving them water in the glass unless they specifically ask water from the glass.
- 4. Work on Building a Rapport with the students:** Just wishing someone good morning or hoping they have a great day may really go a long way in costumer retention strategy.
- 5. Keep Track of Different on Campus Fests and Especially Paradox:** Events like Shaastra, Margazhi, Paradox are times to increase inventory stock exponentially as there is an influx of many people from all over Chennai. This years' sales of CCD for Paradox (unfortunately the data I have, does not record this spike) almost saw a revenue of 1 lakh per day with around 700-1000 walk-ins!

*** Link to the Dataset, all Mathematical Formulae used and Calculations:**

<https://drive.google.com/drive/folders/1YvoP2EDZRYYaZzY0HhEnvcNu-5Vqp-pf>